

MASTER THESIS SUMMARY

**Multi-Sensor Tool Wear Monitoring for High-Precision Turning:
A Machine Learning Approach to Predictive Maintenance**

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1. Motivation

In high-precision CNC turning, cutting tools are subjected to extreme mechanical and thermal loads that lead to inevitable, progressive wear. Left unmonitored, this degradation carries direct consequences for production quality, cost, and process reliability:

- **Quality degradation** – worn tools produce poor surface finish and dimensional inconsistencies in machined parts.
- **Increased costs** – premature tool replacement wastes remaining tool life, while delayed replacement causes scrap and rework.
- **Process disruption** – unexpected tool breakage can damage the workpiece and the machine itself, resulting in costly downtime.

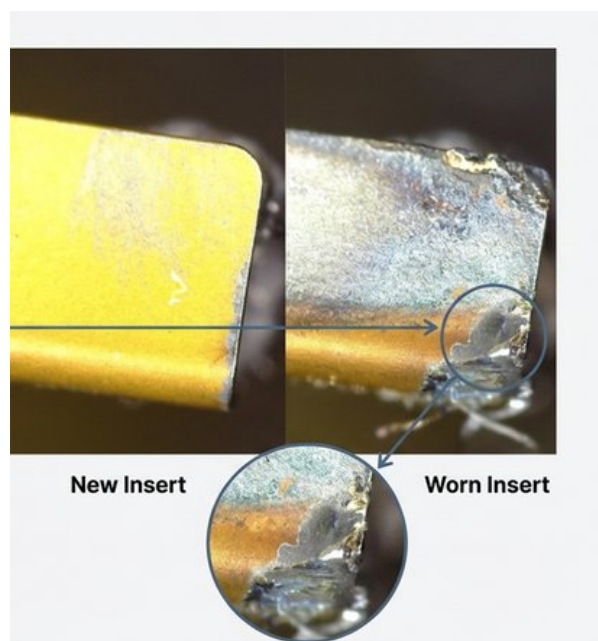


Figure 1. Flank wear progression on a Sandvik-Coromant insert — new (left) versus worn (right).

These risks motivate a shift away from purely reactive or fixed-interval tool replacement strategies toward condition-based, predictive approaches that monitor tool state directly during machining.

2. Proposed Solution

This thesis presents a complete, end-to-end Tool Wear Monitoring (TWM) system that combines a custom multi-sensor hardware platform with machine learning to predict flank wear in real time. The workflow transforms raw physical signals captured during CNC turning into actionable intelligence on tool condition, following five stages:

- **Process** – CNC turning operation is carried out under monitored conditions.
- **Data acquisition** – a custom Tool Holder System (THS) with integrated force, acceleration, and acoustic emission sensors captures live signals.
- **Signal processing & feature extraction** – raw signals are cleaned, segmented, and key features are automatically extracted.

- AI methods – machine learning models (Random Forest and an Artificial Neural Network) are trained on the extracted feature data.
- Prediction – the trained model outputs the tool's condition, predicting flank wear.

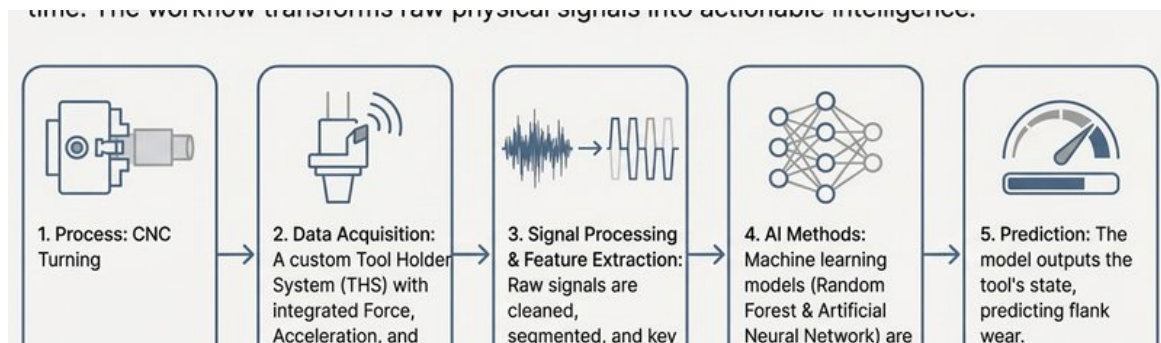


Figure 2. End-to-end workflow of the proposed Tool Wear Monitoring system.

3. Multi-Sensor Tool Holder System — Design

A standard tool holder was redesigned to integrate sensing elements as close to the cutting action as possible. Three conceptual designs for force-sensor integration were modelled and evaluated before selecting a final configuration for manufacturing.

3.1 Sensor selection

Sensor	Specification
Force	Kistler 9132C, piezoelectric, range 0–7 kN
Acceleration	PCB Piezotronics J352C68, single-axis, range $\pm 491 \text{ m/s}^2$
Acoustic emission	Vallen Systeme VS75V, range 30–120 kHz

3.2 Design validation

Static structural simulations in ANSYS were used to evaluate the conceptual designs. A 200 N cutting force was applied to the tool tip to analyse the resulting load transferred to the force sensor. The first conceptual design showed the most favourable load transfer and was selected for manufacturing.

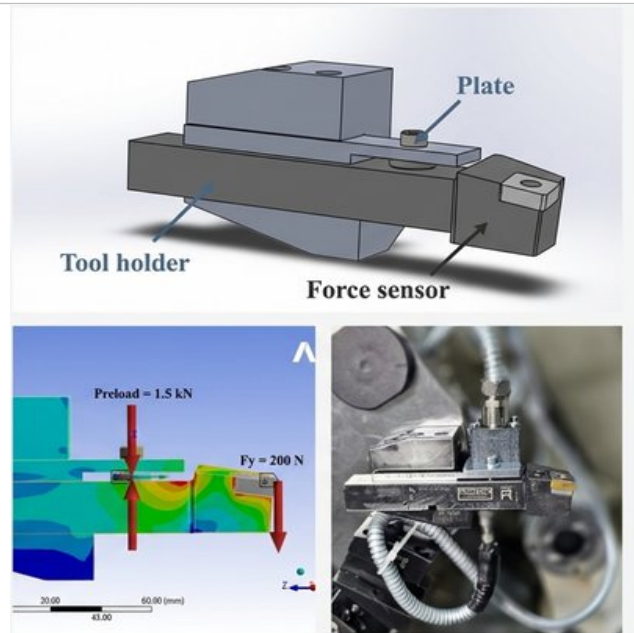


Figure 3. Conceptual force-sensor integration (top) and finite-element validation of the tool holder plate at $F_y = 200\text{ N}$ (bottom).

4. Multi-Sensor Tool Holder System — Final Build

The final manufactured Tool Holder System successfully integrates all four selected sensors within a purpose-built protective housing:

- **Force sensor** – integrated between the tool holder and a custom-designed mounting plate.
- **Acceleration sensors** – two sensors measure vibration in both the cutting and feed directions.
- **Acoustic emission sensor** – attached directly to the machine turret via a magnetic clamp.
- **Sensor and cable protection** – a custom 3D-printed (FDM) enclosure protects the force sensor, while a plastic covering hose shields its cable from sharp chips.

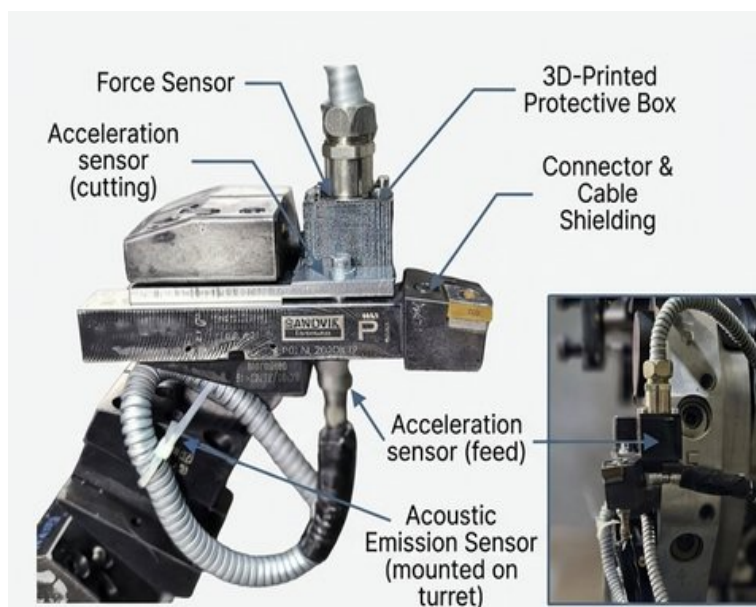


Figure 4. Final Tool Holder System with integrated force, acceleration, and acoustic emission sensors.

5. Experimental Setup

A series of sixteen turning tests was performed on 42CrMo4 steel using four Sandvik-Coromant cutting inserts, with the multi-sensor THS mounted on a CNC turning machine.

- **Wear measurement** – after each turning pass, a Dino-Lite digital microscope fixed inside the machine captured high-resolution images of the cutting edge. Flank wear (VB) was measured directly from these images, providing the ground-truth data for training the machine learning models.
- **Data acquisition** – signals from all four sensors were captured simultaneously — force and acceleration sampled at 48 kHz, acoustic emission at 192 kHz — passed through dedicated amplifiers and conditioners, and digitised by a QuantumX data acquisition device.

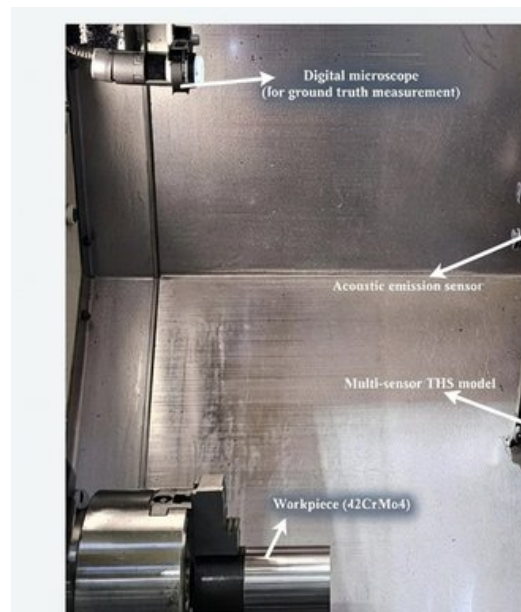


Figure 5. CNC turning test cell with the multi-sensor THS mounted on the workpiece (42CrMo4 steel).

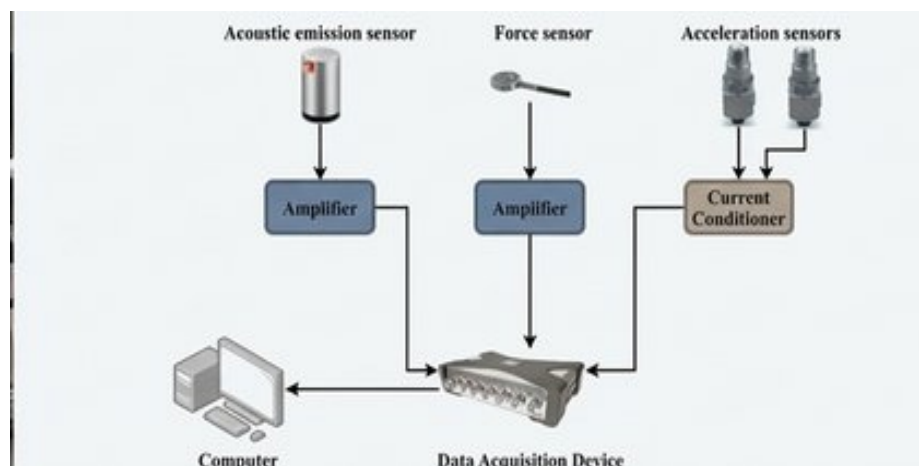


Figure 6. Signal-conditioning and data-acquisition chain for the four sensor channels.

6. Signal Processing

The raw data stream contains periods of inactivity and noise. To ensure the machine learning models train only on relevant information, a two-step processing pipeline was applied:

- **Segmentation** – a peak-finding algorithm (SciPy find_peaks) automatically identifies the start and end of each cutting pass, trimming the signal to include only the active machining period.
- **Filtering** – a Butterworth high-pass filter (10 Hz cutoff) is applied to the cutting-force signal to remove low-frequency drift caused by thermal effects on the sensor.

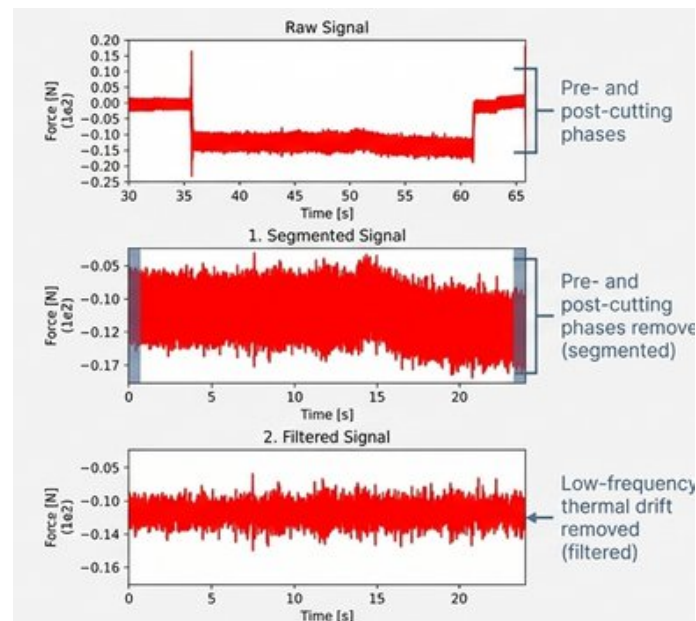


Figure 7. Raw force signal before and after segmentation and high-pass filtering.

7. Machine Learning Approach

Two machine learning models were trained and evaluated on the same dataset, derived from the sixteen full turning tests:

- **Random Forest (RF)** – a robust ensemble learning method based on decision trees.
- **Artificial Neural Network (ANN)** – a feed-forward neural network with five hidden layers, designed to capture complex non-linear relationships in the sensor data.

To quantify the value of multi-sensor fusion, model performance was evaluated across three distinct feature sets, each incorporating an additional sensor modality:

Feature Set	Sensors Included
Feature Set 1	Force + Acceleration (Cutting)
Feature Set 2	Force + Acceleration (Cutting) + Acceleration (Feed)
Feature Set 3	Force + Acceleration (Cutting) + Acceleration (Feed) + Acoustic Emission

8. Conclusion

This work successfully developed and validated a complete workflow for intelligent tool wear monitoring, encompassing:

- A robust, custom-designed multi-sensor hardware system (THS).
- An automated data-processing pipeline capable of handling real-world sensor signals.

- An optimised ANN model that accurately predicts tool wear by fusing data from multiple sensors.

Together, these contributions support a shift from reactive maintenance — scheduled changes and unexpected failures — toward predictive maintenance, enabling data-driven decisions, optimised tool life, and enhanced part quality.

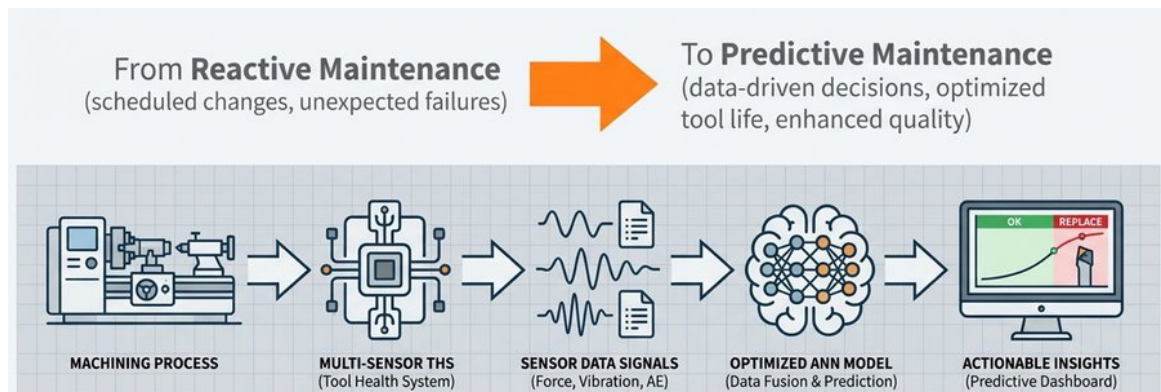


Figure 8. From reactive to predictive maintenance: the validated end-to-end monitoring system.

Appendix A. Engineering Drawings

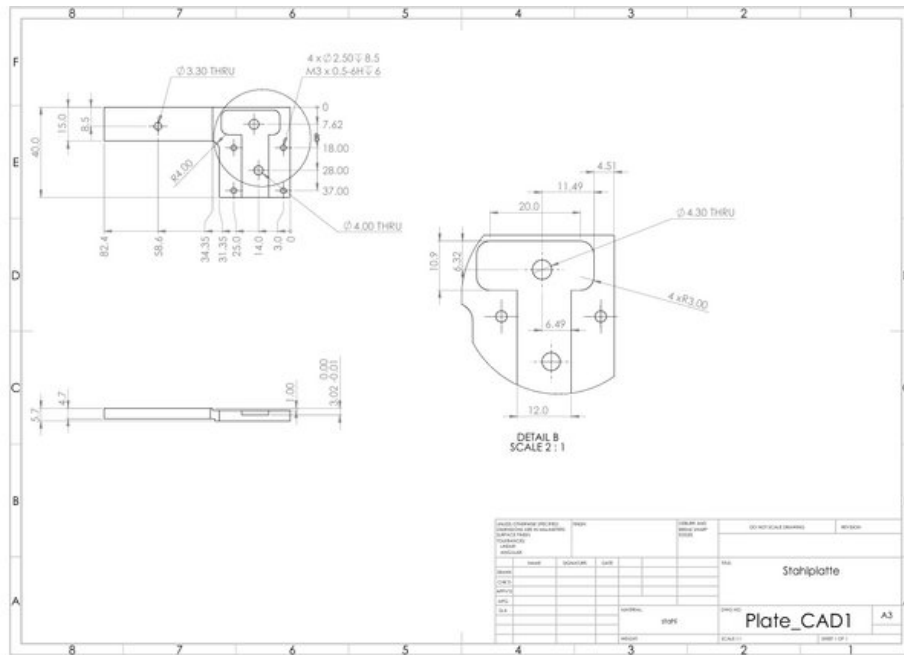


Figure A.1. Detail drawing of the custom sensor mounting plate (Plate_CAD1).

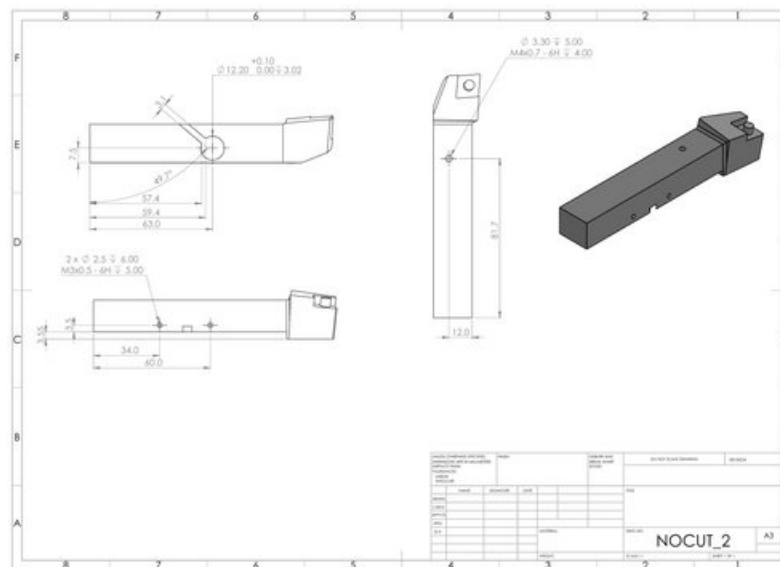


Figure A.2. Detail drawing of the modified tool holder (NOCUT_2).